

**Sardar Vallabhbhai National Institute of Technology
(SVNIT) Surat
Department of Artificial Intelligence
5th Sem , B.Tech. (AI) (2026-27)**

**Information Security and Cryptography - AI 331
Lab Assignment 5**

Scenario

A ground control station needs to securely transmit a short tactical command to a field drone using a 2×2 Hill Cipher.

- **Key Matrix:** $K = \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix}$

- **Plaintext:** MEET AT BASE

Tasks

Write a program to:

1. Construct and display the 2×2 Hill Cipher key matrix.
2. Convert the plaintext to uppercase, remove spaces, and map letters to numbers using **A = 0, B = 1, ..., Z = 25**.
3. Divide the plaintext into valid **2-letter blocks**.
4. Encrypt each block using the Hill Cipher formula.
5. Display the encrypted numerical blocks and the corresponding **ciphertext**.
6. Find the **modular inverse of the key matrix modulo 26**.
7. Decrypt the ciphertext.
8. Convert the decrypted numerical values back to letters and **verify that the original plaintext is recovered**.